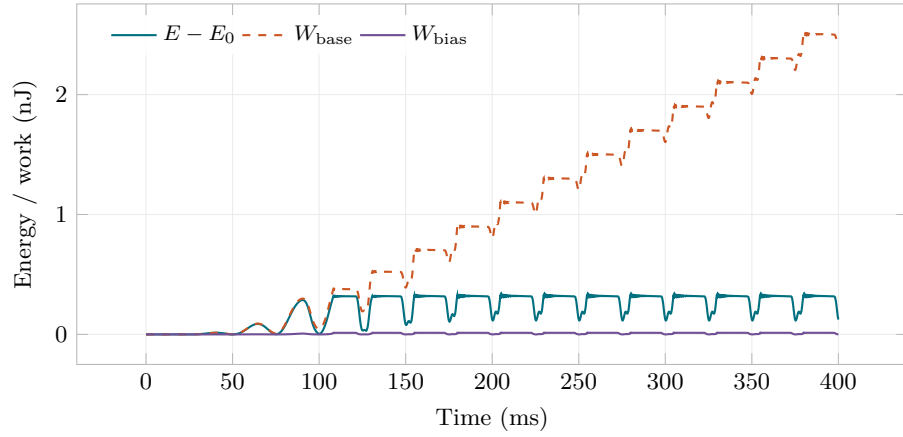


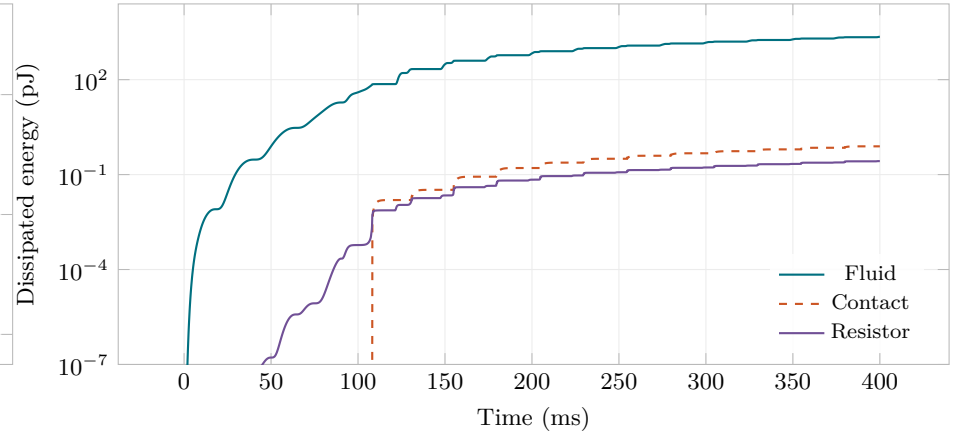
Energy accounting and a separate contact probe

Panels (a)–(c): driven run Panel (d): unforced 600 μs initial-condition experiment

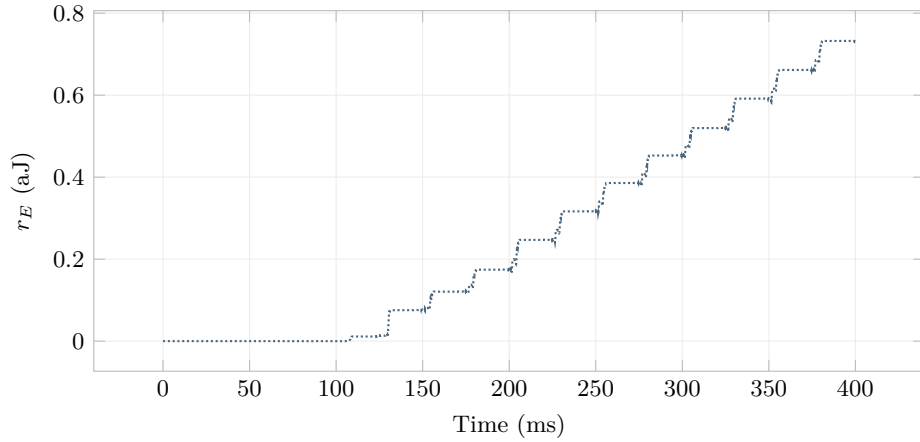
(a) Stored energy and source work



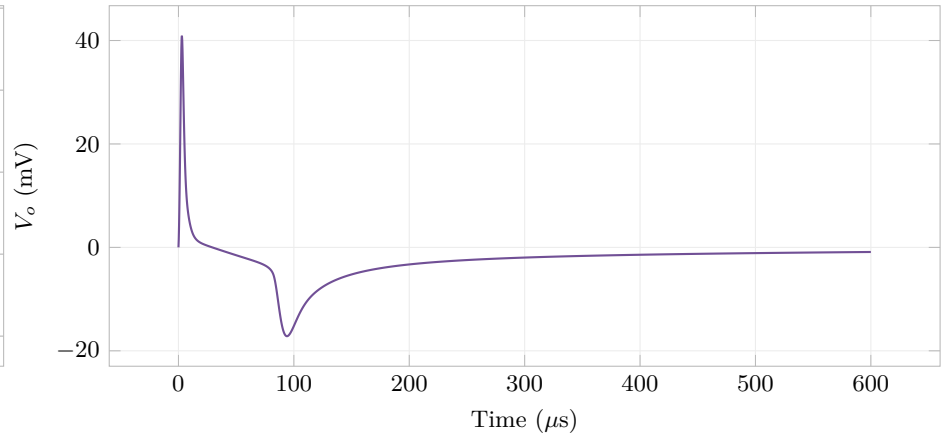
(b) Cumulative dissipation



(c) Energy-balance residual



(d) Unforced contact-probe voltage



$$r_E = E - E_0 - W_{\text{base}} - W_{\text{bias}} + E_R + D_{\text{film}} + D_{\text{wall}} + D_{\text{struct}}; 1 \text{ aJ} = 10^{-18} \text{ J}.$$

The larger signal in (d) follows a specified near-contact initial displacement and velocity; it is not the driven steady-state voltage. Numerical energy closure is not experimental validation.